

Audio-Driven Violin Performance Animation with Clear Fingering and Bowing

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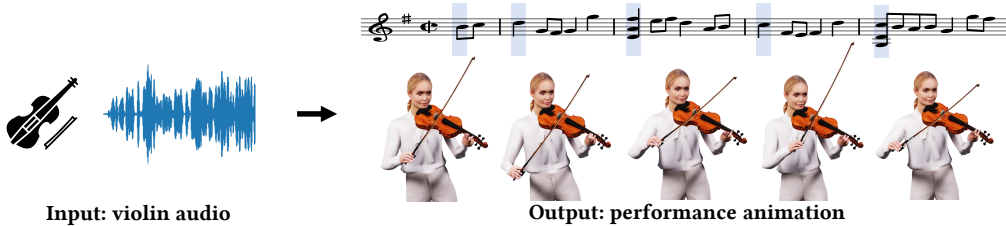


Figure 1: A violin performance animation synthesized by our method (violinist 3d model from renderpeople.com).

ABSTRACT

This paper presents an audio-to-animation synthesis method for violin performance. This new approach provides a fine-grained violin performance animation using information on playing procedure consisting of played string, finger number, position, and bow direction. We demonstrate that our method is capable of synthesizing natural violin performance animation with fine fingering and bowing through extensive evaluation.

CCS CONCEPTS

• Computing methodologies → Motion processing; • Applied computing → Media arts; Sound and music computing.

KEYWORDS

Motion generation, music information retrieval, neural networks.

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1 INTRODUCTION

Automatic body motion generation from audio for musical performance has attracted increasing attention recently. This technique enables anyone to easily create musical performance animations for their favorite audio without special devices such as motion capture

suits. To generate body motion for violin performance, [Shlizerman et al. 2018] and [Kao and Su 2020] directly estimated the sequence of body joint positions from audio features, while [Shrestha et al. 2022] proposed a motion refinement method based on audio features for those results. These were pioneering works in trying to solve this task in an end-to-end manner. However, it is still challenging to create adequate motion due to the ambiguous mapping between observed audio features and joint positions, as musical performance requires precise motion to produce the target sound.

In this paper, we propose an audio-to-motion synthesis method that leverages the playing procedure information. In contrast to the studies above, we tackle this problem by estimating the playing procedure information for hands from audio features prior to motion synthesis. Specifically, we estimate the played string, finger number, position (*i.e.*, where on the fret to position the left hand), and bow direction, and then procedurally determine the performance motions. In this approach, the process of producing sound is faithfully modeled from the perspective that the observed audio features are generated in accordance with actions based on a particular playing procedure. The necessary information is far less complicated than joint positions and thus should be easier to predict, since the number of possible playing procedures is limited to just a few combinations of discrete features when conditioned by pitch and timbre, whereas each procedure has infinite possible joint positions in a continuous space. We show that our method is capable of synthesizing natural motion with accurate playing procedure even for the left hand and fingers other than the right hand, which was addressed in our previous work [Hirata et al. 2021].

2 METHOD

Our method consists of three components (Figure 2), a playing procedure model, a hand motion synthesizer, and a body motion model. The playing procedure model estimates four hand-related features consisting of the played string, the finger number, the position, and the bow direction. The hand motion synthesizer generates the left hand motion from the string, the finger, and the position features

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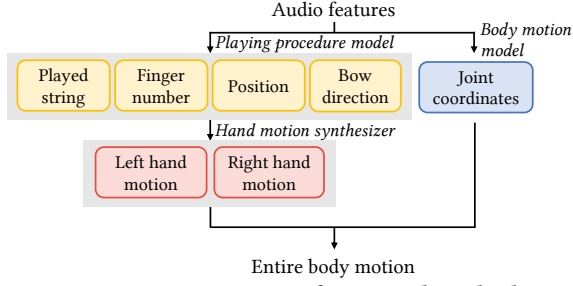


Figure 2: An overview of proposed method.

and the right one from the string and the bow features. The body motion model estimates the 3D coordinates for the entire body.

The playing procedure model consists of two networks with the same architecture consisting of a one-layer long short-term memory (LSTM) followed by a fully connected layer. Both take as an input of a short-time Fourier transform (STFT) of a violin playing sound but output different playing procedure information. Specifically, one of the two networks predicts the played string $\hat{L}^{str} = \hat{l}_{1:T}^{str} \in \{0, 1, 2, 3\}^T$ and the finger number $\hat{L}^{fing} = \hat{l}_{1:T}^{fing} \in \{0, 1, 2, 3, 4\}^T$ simultaneously, where T is the number of total time frames. The string numbers $\{0, 1, 2, 3\}$ correspond to $\{E, A, D, G\}$ strings and the finger numbers $\{0, 1, 2, 3, 4\}$ correspond to $\{\text{open string (i.e., no finger is pressed down), index, middle, ring, pinky}\}$ fingers, respectively. Given $\hat{L}^{str}$, $\hat{L}^{fing}$, and the pitch (i.e., the input audio), we can uniquely decide the position $\hat{L}^{pos} = \hat{l}_{1:T}^{pos} \in \{1, 2, \dots, 12\}^T$, a 12-level indicator of where to position the left hand on the fret. The other network predicts the bow direction at each time frame $\hat{L}^{ud} = \hat{l}_{1:T}^{ud} \in \{0, 1\}^T$ (0:down, 1:up), the same as in [Hirata et al. 2021]. The playing procedure information is finally refined by majority voting within each segment divided based on the onset time of the input audio.

The hand motion synthesizer determines the rotation for each hand joint by inverse kinematics (IK) to meet the estimated playing procedure information. We set IK chains for each finger of the left hand, for the left arm, and for the right arm. The target positions for the fingers are determined from $\hat{L}^{str}$, $\hat{L}^{fing}$, and $\hat{L}^{pos}$ to press the corresponding position on the fret, and for the left arm from $\hat{L}^{pos}$. For the right arm, the target position is determined from $\hat{L}^{str}$ and $\hat{L}^{ud}$ to track on the trajectory corresponding to each of the four strings. The IK is solved at the frames corresponding to each onset time, and the joint parameters between them are interpolated.

The body motion model outputs the sequence of the 3D coordinates for body joints on the basis of mel-frequency cepstral coefficient input. For this model, we adopt the LSTM-based network used in a prior method [Shlizerman et al. 2018]. This component synthesizes motions aside from the hand-related ones. The final animation is created by applying the synthesized motion to corresponding body parts and fixing the violin body to the left shoulder and the bow to the right wrist.

3 EVALUATION

We evaluated our results quantitatively and qualitatively. To train our model, we constructed a new dataset consisting of one hour (30 pieces) of violin performance by a single player, which provides

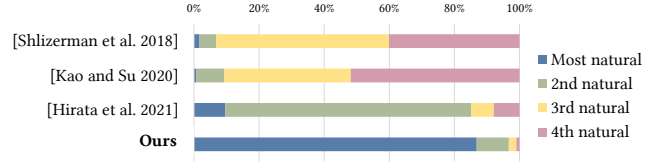


Figure 3: Subjective evaluation for naturalness.

paired audio, joint coordinates from video, and aligned playing procedure information. A leave-one-out scheme was performed for data splitting to ensure that the same piece was not shared with other subsets. Audio features were extracted with [McFee et al. 2015]. To evaluate whether the final output accurately corresponds to the input audio in terms of temporal synchronization and playing procedure accuracy, we determined the F-measure for the onset and the accuracies for the playing procedure information. Note that the onset needs to be evaluated because it is not necessarily detected at the start time of each musical note. The results showed that the F-measure for the onset at tolerances of 0.033, 0.067, and 0.100 seconds were 0.582, 0.867, and 0.913 and the accuracies for the played string, the finger number, the position, and the bow direction were 0.965, 0.879, 0.892, and 0.820, respectively.

We also conducted a subjective evaluation with 30 participants to compare our method with existing ones. We randomly chose ten excerpts from our dataset and synthesized the animation with each method. The participants were instructed to rank the naturalness of the synthesized animation. The results are shown in Figure 3, where we can see that our results are far more natural than the others. We also illustrate our result in Figure 1. Please see our supplementary video for more detailed examples of the animation.

4 CONCLUSION AND FUTURE WORK

In this paper, we proposed a new motion synthesis approach for violin performance from raw audio and demonstrated that it can synthesize natural motion even for fingers. In future work, we plan to expand our dataset to multiple performers for more versatility.

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